

Transition of NBA Game Strategy After the Introduction of the 3-Point Shot

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1 Introduction

Modern basketball is a sport that is extensively analyzed at a statistical level. The following proposes to examine the impact of the three-point shot, introduced to the NBA in the 1979-1980 season, on the evolution of game strategies and competitive results. The analysis highlights the transition toward a new game paradigm, in which teams have shifted from a marginal use of the three-point shot to its massive optimization. To quantify this change, the effective efficiency metric (*Effective Field Goal Percentage*, *eFG%*), defined in the literature by Dean Oliver [1], will be used, allowing us to weigh the value of shots based on their actual scoring output. Furthermore, this work will extend the investigation to error and possession management, analyzing how variables such as turnovers and offensive rebounds influence win probability. Finally, the analysis focuses on how the advent of the three-point shot has altered the management of these parameters, imposing a new strategic balance between offense and defense.

2 Methods

The data used in this analysis were extracted from a public dataset in SQLite format [2]. The processing pipeline was divided into three main phases:

1. Acquisition and Preprocessing: Through SQL queries, the dataset relating to individual games was isolated. Data cleaning was performed in Python using the pandas library, removing incomplete records and filtering the time interval 1985-2022, as earlier seasons presented issues and missing values in the available data.
2. Exploratory and Statistical Analysis: For the preliminary inspection of variables, the D-Tale tool [3] was employed, which easily visualizes and analyzes pandas data structures. Subsequently, the validation of linear models and the calculation of Pearson correlation coefficients (r), along with their respective p values, and the generation of linear regression models were evaluated using `scipy.stats`.

3. Visualization: The graphical representation of the results was created using the matplotlib library.

The source code is available in the public repository [4].

3 Results

The analysis of the results begins with an examination of the evolution of offensive strategy. Figure 1 illustrates the NBA's systematic transition towards an intensive use of the 3-point shot; a steady growth in shot volume since 1985 is observed. However, this increase in shot volume does not translate into a direct increase in win probability ($r \approx 0.09$ with $p_{value} \approx 0.005$), suggesting that the volume of long-distance shots is an insufficient variable to describe competitive success.

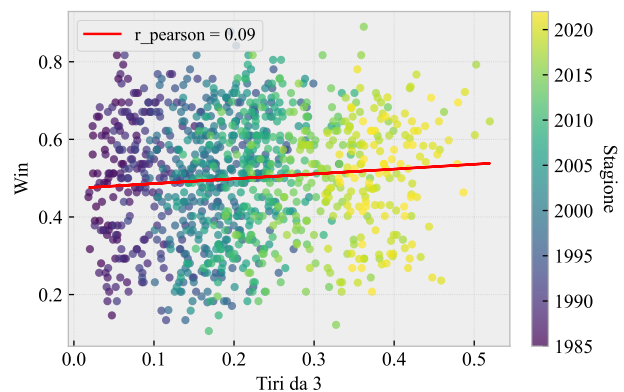


Figure 1: Win rate for each individual team as a function of the 3-point shot rate for the seasons spanning 1985 to 2022. The data show a clear transition in the playing style of NBA games, described by a growing seasonal migration towards the use of 3-point shots. However, there are no direct correlations between 3-point shot volume and winning games.

To overcome the limitations of the volumetric data and properly evaluate a team's efficiency, the effective efficiency metric (*Effective Field Goal Percentage*, *eFG%*)

[1]) was introduced, defined as:

$$eFG\% = \frac{FGM + 0.5 \times FG3M}{FGA}. \quad (1)$$

In this formulation, FGM represents total field goals made, FG3M represents three-point field goals made, and FGA stands for total field goals attempted. The 0.5 corrective factor applied to three-pointers acts as a "weight" to compensate for the higher value of a 3-point shot compared to a 2-point shot. In terms of efficiency, Equation 1 reflects the fact that a three-point shot is worth 50% more than a two-point attempt.

The introduction of $eFG\%$ resolves the paradox observed previously. As shown in Figure 2, the correlation between effective efficiency and the win rate (Win) is extremely solid, with a Pearson coefficient of $r = 0.48$ and clear statistical significance ($p_{value} < 0.001$). Moving from $r \approx 0.09$ (volume) to $r \approx 0.48$ (efficiency) demonstrates that winning is not correlated to *how much* a team shoots from three, but to *how well* they do it in relation to total attempts. This leap in the model's predictive capability validates $eFG\%$ as a primary driver of offensive performance in the modern NBA.

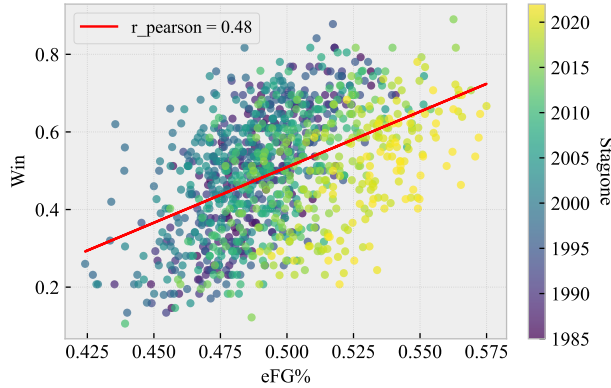


Figure 2: Win rate for each individual team as a function of effective efficiency ($eFG\%$) for the seasons spanning 1985 to 2022. A solid and statistically significant positive correlation is observed. This result validates $eFG\%$ as a driver of competitive success, demonstrating that winning is tied to execution quality rather than the number of attempts made.

The investigation was then extended to "error management", examining turnovers (*Turnover Percentage*, $TOV\%$ [1]) and offensive rebounds (*Offensive Rebound Percentage*, $ORB\%$ [1]), defined as:

$$TOV\% = \frac{TOV}{FGA + 0.44 \times FTA + TOV} \quad (2)$$

$$ORB\% = \frac{ORB}{ORB + Opp_{DRB}} \quad (3)$$

In the previous equations, TOV represents total turnovers, FGA represents total field goals attempted, and FTA stands for free throws attempted. The 0.44 coefficient in Eq. 2 is a corrective statistical factor used to estimate the number of possessions resulting from free throws, accounting for situations like and-ones or technical fouls. In Eq. 3, ORB indicates the offensive rebounds captured by the analyzed team, while Opp_{DRB} represents the defensive rebounds obtained by the opponents.

While Eq. 2 quantifies possession dissipation (the error that ends a play without a shot), Eq. 3 measures the team's ability to recycle a missed shot, regenerating a new possession. Data analysis highlights a fundamental asymmetry between these two types of "misses".

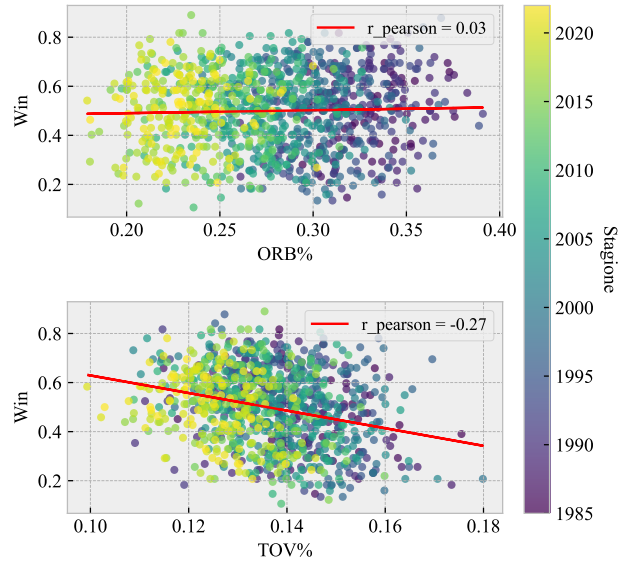


Figure 3: Comparative analysis of possession management through the win rate as a function of $ORB\%$ and $TOV\%$ (1985-2022). For offensive rebounds ($ORB\%$), we observe an almost zero correlation, and the lack of statistical significance ($p_{value} \approx 0.29$) confirms the metric's irrelevance in determining ultimate success. For turnovers ($TOV\%$), a clear negative trend is detected, confirming that ball protection and minimizing possession dissipation are critical variables for increasing win probability.

As illustrated in Fig. 3, $TOV\%$ shows a statistically robust negative correlation ($r \approx -0.27$ and $p_{value} < 0.001$), confirming that ball protection is a critical performance driver. Conversely, $ORB\%$ does not present a statistically significant correlation with winning ($r \approx 0.03$ and $p_{value} \approx 0.29$). This seemingly counterintuitive phenomenon (since a "second chance" should theoretically increase the probability of scoring) might have a physical and tactical explanation in the *spacing* generated by the three-point shot. It can be hypothesized that longer

shots generate more unpredictable rebounds; teams seem to abandon offensive rebounding to ensure a solid defensive transition, thereby minimizing the risk of suffering fast breaks.

Metric	Correlation vs Season
Volume (3-Pt Shots)	0.89 ($p < 0.001$)
Efficiency ($eFG\%$)	0.51 ($p < 0.001$)
Turnovers ($TOV\%$)	-0.42 ($p < 0.001$)
Rebounds ($ORB\%$)	-0.82 ($p < 0.001$)

Table 1: Correlation analysis (Pearson’s r) between game metrics and temporal evolution (1985-2022).

To better highlight seasonal trends, a further correlation analysis was conducted between game metrics and the time variable (Tab. 1). While the relationship between efficiency ($eFG\%$) and wins remains a technical constant of the game across eras, a clear transition in performance is observed. The link between $eFG\%$ and advancing seasons ($r \approx 0.51$) suggests that the minimum efficiency standards required to be competitive have risen drastically over the last forty years. By way of interpretation, this increase in $eFG\%$ over time can be traced back to a synergy of evolutionary factors. On one hand, tactical optimization: the conscious choice to favor the three-point shot due to its higher mathematical value. On the other hand, a possible rise in the average shooting talent of players, who are now exposed to perimeter technical training much earlier and more specifically than in previous decades. Finally, another interesting finding concerns the collapse of offensive rebounds ($r \approx -0.82$ relative to the season). This trend can be interpreted as a strategic mutation of the "modern NBA"; the increase in *spacing*, as hypothesized previously, has made recovering the ball after a miss a more uncertain and risky event, pushing teams to prefer the stability of a defensive transition over the pursuit of a second opportunity. In this sense, possession protection (a decrease in $TOV\%$ with $r \approx -0.42$) has become the true management priority to minimize the dissipation of offensive opportunities.

4 Conclusions

The conducted study allows us to outline the drivers of success in the modern NBA, distinguishing between the effectiveness of individual strategies and their historical evolution. The analysis confirmed that the transition toward perimeter shooting represents a structural migration of playing style; however, the weak correlation between three-point shot volume and wins ($r \approx 0.09$) suggests that success does not lie in a mere increase in attempts, but in execution quality. The shift to the effective efficiency met-

ric ($eFG\%$) highlighted a solid correlation ($r \approx 0.48$) with wins, demonstrating that competitive advantage is dictated by the ability to convert the high potential of the three-point shot into actual baskets. It is nevertheless fundamental to observe, as highlighted by Tab. 1, that efficiency itself has systematically grown over time ($r \approx 0.51$ relative to the season).

Regarding possession management, ball protection ($TOV\%$) is confirmed as a pillar of system stability ($r \approx -0.27$), indicating that minimizing the waste of offensive opportunities is fundamental to maintaining a winning record. Conversely, possession regeneration through offensive rebounds ($ORB\%$) proved to be an irrelevant variable in determining success, also showing the sharpest historical decline of the entire analysis ($r \approx -0.82$ with the season).

Ultimately, the modern NBA’s playing style favors scoring efficiency and possession security over the pursuit of second chances. The *spacing* generated by the three-point shot appears to have redefined game strategy, revealing a team preference for ensuring defensive transition solidity rather than risking precious possessions in rebounding battles. In conclusion, winning in modern basketball is a balance between optimizing shot value and minimizing the waste of offensive opportunities.

References

- [1] D. Oliver, “Dean Oliver’s Four Factors Revisited,” <https://www.researchgate.net/publication/370949495>, 2023, [Online; visitato il 28-Apr-2026].
- [2] W. Walsh, “NBA Database,” <https://www.kaggle.com/datasets/wyattowalsh/basketball>, 2024, [Online; visitato il 25-Apr-2026].
- [3] M. A. Technology, “Libreria Python D-tale,” <https://pypi.org/project/dtale/>.
- [4] G. Di Placido, “NBA Strategies Analysis,” <https://github.com/giuseppe-dp/nba-strategies-analysis>, 2026, [GitHub Repository].